Compiler Optimization Report - YOLOv8s Object Detection

June 2026

Document Version 3.0
SDK Version v26.05

1 Summary

For YOLOv8s Object Detection at 1088x1920x3 resolution:

Performance simulation estimates a Total Processing Time (for the compiled ONNX file) of

- 17.26 ms, of which
- 6.23 ms is Analog NPU processing time, which corresponds to
  - 75.0% ACE utilization, using 24 Analog Compute Engines; and of which
- 11.03 ms is Digital NPU processing time,
  - using only 4 Digital NPU cores, which is sufficient to achieve 30 fps.

The number of used Digital NPU cores (4) was selected to achieve at least 30 fps. If many more digital NPU cores were used, the Digital NPU processing time could be substantially shortened. For example, 100 Digital NPU cores could reduce the Digital NPU processing time by ~25x, resulting in a total (Analog + Digital) processing time of ~7 ms, corresponding to ~150 fps.

There are no digital bottlenecks in the Analog NPU processing time.

Automatic optimization performs as well as the best manual optimization.

We may be able to improve performance in the future by finding a more optimal placement of weight blocks to ACEs (to increase the ACE utilization), but note that the current ACE utilization of 75.0% is already very high.

We may be able to improve performance in the future by allowing weight block widths to exceed 256. The hardware design recently increased support to 512, and the compiler will add support for 512 in the future.

The estimated power is

- 0.3 Watts at the target frame rate (30 fps), and
- 0.6 Watts at the maximum frame rate (57 fps), if restricted to using only 4 Digital NPU cores.

Power estimates are approximate, and will be more precise in the future.

2 Reproducing the results

The results in this report were produced using the `“high_optimization”` compiler config for YOLOv8s Object Detection.

Note that some variance in results is expected (up to a few percent), due to non-deterministic, multi-threaded compiler optimizations.

The `mythic-compiler` command was:

```
mythic-compiler \
    --input-artifact $MYTHIC_SDK_ROOT/models/training/yolov8/yolov8s_trained.tar.gz \
    --compiler-config $MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolov8/compiler/
yolov8s_1088x1920x3_m2024_high_optimization.yaml \
    --output-artifact /rscratch/tonbomythic3/<your_username>/
yolov8s_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

The `mythic-ppa-estimators` command for the maximum frame rate was:

```
mythic-ppa-estimators \
    --estimate-performance \
    --estimate-power \
    /rscratch/tonbomythic3/<your_username>/
yolov8s_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

The `mythic-ppa-estimators` command for the target frame rate (30 fps) was:

```
mythic-ppa-estimators \
    --estimate-performance \
    --estimate-power \
    --power-inference-rate 30 \
    /rscratch/tonbomythic3/<your_username>/
yolov8s_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

3 Performance estimates

The `mythic-ppa-estimators` tool outputs the following performance statistics:

```
Analog NPU Total Estimated Processing Time for Compiled ONNX File: 6.23 ms
Analog NPU Estimated Frame Rate for Compiled ONNX File: 160.54 fps (frames per second)
Critical Path ACE Latency: 6.23 ms
Maximum SRAM Read/Write Time (over 6 ACE tiles): 5.51 ms
Maximum SIMD Operation Time (over 6 ACE tiles): 0.00 ms
Average SRAM Read/Write Time (averaged over 6 ACE tiles): 4.37 ms
Average SIMD Operation Time (averaged over 6 ACE tiles): 0.00 ms
Total Executed ACE Operations: 700,740
Total Executed ACE MACs: 106,348,953,600
Maximum Theoretical ACE Execution Time (With No Parallelization Using 1 ACE): 112.12 ms
Minimum Theoretical ACE Execution Time (With Even Parallelization Across 24 ACEs): 4.67
 ms
ACE Utilization (Minimum Theoretical Time/Estimated Processing Time): 75.00%
Total SRAM Bytes Read Across Chip: 1,849,990,648 bytes
Total SRAM Bytes Written Across Chip: 980,385,584 bytes
Estimated Die Area to Achieve Estimated Processing Time: 126.55 mm^2
-------------------------------------------------------------------
Digital NPU Performance Metrics (from separate ONNX graph processing)
  Total Estimated MACs Targeting Digital NPU: 22,000,000
  Digital NPU MAC Utilization: 0.78%
  Digital NPU Cores: 4
  Digital NPU Frequency: 1,000 MHz
  Digital Estimated Frame Processing Time: 11.03 ms
  Digital Estimated Frame Rate: 90.69 fps (frames per second)
-------------------------------------------------------------------
```

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```
Combined Analog + Digital NPU Latency: 17.26 ms
Combined Analog + Digital NPU Total Estimated Frame Rate: 57.95 fps (frames per second)

NOTE: The information reported above may include the Analog NPU or the Analog
NPU and a digital NPU as indicated. This estimated processing time and
associated frame rate only covers the execution of the compiled ONNX model(s)
running on one or both of those compute solutions in isolation. It does not
include PCIe transfer overhead. Performance estimation for additional processing
steps and a combined compute hardware solution will be added in future versions.
```

4 Power estimates

The `mythic-ppa-estimators` tool outputs the following power estimates.

Power at the target frame rate (30 fps):

```
Process Nodes:  Analog 28nm, Digital 5nm
Number of ACEs: 24
Number of Digital NPU Cores: 4
Inference Rate: 30 frames/second (fps)

Analog NPU Estimated Power: 0.311 W for target ONNX file
    Functional Unit Power: 0.253 W
    Interconnect Power:    0.059 W

Digital NPU Estimated Power: 0.014 W for digital target ONNX file

Total Combined (Analog + Digital NPU) Power: 0.325 W

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future
versions.
```

Power at the maximum frame rate (57 fps):

```
Process Nodes:  Analog 28nm, Digital 5nm
Number of ACEs: 24
Number of Digital NPU Cores: 4
Inference Rate: 57 frames/second (fps)

Analog NPU Estimated Power: 0.534 W for target ONNX file
    Functional Unit Power: 0.423 W
    Interconnect Power:    0.112 W

Digital NPU Estimated Power: 0.026 W for digital target ONNX file
```

```
Total Combined (Analog + Digital NPU) Power: 0.560 W

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future
versions.
```

5 Compiler reports

5.1 L0 IR stats

From near the end of mythic-compiler stdout:

```
================================= L0 IR stats ==================================
                   Tiles: 7
                 Buffers: 169
               Launchers: 266
           ACE Launchers: 107
             ACE Weights: 18620528
        ACE calculations: 700740
    Max ACE calculations: 121040 on tile 4_2
================================================================================
```

5.2 ACE utilization report

The compiler-generated `ace_utilization.txt`:

```
Summary:

  - 80.92% ACE utilization is predicted by the compiler,

    - assuming no digital bottlenecks (which can typically be eliminated).

  - 700,740 ACE calculations are spread across 24 ACEs, which is

    - 29,197.50 ACE calculations per ACE (on average).

  - 36,084 ACE calculations are in the critical path, implying

    - 5.77 ms predicted inference time (160 ns per ACE calculation).

+------+------+--------------+--------------+-------------------+-------------+
|      |      |              |      ACE     |     Predicted     |  Predicted  |
|      |      |      ACE     | calculations |    Full Network   |     ACE     |
| Tile | ACEs | calculations |    Per ACE   | ACE Critical Path | Utilization |
+------+------+--------------+--------------+-------------------+-------------+
|  1_1 |    0 |          N/A |          N/A |               N/A |         N/A |
|      |      |              |              |                   |             |
|  3_1 |    4 |      120,632 |    30,158.00 |            36,084 |      83.58% |
|      |      |              |              |                   |             |
|  4_1 |    4 |      112,166 |    28,041.50 |            36,084 |      77.71% |
|      |      |              |              |                   |             |
|  1_2 |    4 |      119,680 |    29,920.00 |            36,084 |      82.92% |
|      |      |              |              |                   |             |
|  2_2 |    4 |      109,446 |    27,361.50 |            36,084 |      75.83% |
|      |      |              |              |                   |             |
|  3_2 |    4 |      117,776 |    29,444.00 |            36,084 |      81.60% |
|      |      |              |              |                   |             |
|  4_2 |    4 |      121,040 |    30,260.00 |            36,084 |      83.86% |
+------+------+--------------+--------------+-------------------+-------------+
|  All |   24 |      700,740 |    29,197.50 |            36,084 |      80.92% |
+------+------+--------------+--------------+-------------------+-------------+
```

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5.3 Weight utilization report

The compiler-generated `weight_utilization.txt`:

```
Summary:

  - 62.31% of analog weight storage capacity is used, using

  - 18,620,528 weights out of 29,884,416 capacity, using

  - 107 weight blocks.

+------+--------+------------+------------+-------------+
|      | Weight |  Allocated |   Weight   |             |
| Tile | Blocks |   Weights  |  Capacity  | Utilization |
+------+--------+------------+------------+-------------+
|  3_1 |     17 |  2,661,952 |  4,980,736 |      53.44% |
|      |        |            |            |             |
|  4_1 |     15 |  2,579,712 |  4,980,736 |      51.79% |
|      |        |            |            |             |
|  1_2 |     18 |  3,741,456 |  4,980,736 |      75.12% |
|      |        |            |            |             |
|  2_2 |     18 |  3,284,160 |  4,980,736 |      65.94% |
|      |        |            |            |             |
|  3_2 |     21 |  3,423,136 |  4,980,736 |      68.73% |
|      |        |            |            |             |
|  4_2 |     18 |  2,930,112 |  4,980,736 |      58.83% |
+------+--------+------------+------------+-------------+
|  All |    107 | 18,620,528 | 29,884,416 |      62.31% |
+------+--------+------------+------------+-------------+
```

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5.4 SRAM utilization report

The compiler-generated `sram_utilization.txt`:

```
Summary:

   - 5,168,651 bytes of SRAM is used, consisting of

     - 4,826,112 bytes for buffers (93.37%) and

     - 342,539 bytes for control flow (6.63%), using

   - 20.54% of the simulated on-chip SRAM capacity.

+------+------------+--------------+------------+------------+-------------+
|      |   Buffer   | Control Flow |   Total    |    SRAM    |             |
|      | Allocation |  Allocation  | Allocation |  Capacity  |             |
| Tile |  (bytes)   |   (bytes)    |  (bytes)   |  (bytes)   | Utilization |
+------+------------+--------------+------------+------------+-------------+
|  3_1 |    531,968 |       59,324 |    591,292 |  4,194,304 |      14.10% |
|      |            |              |            |            |             |
|  4_1 |    314,176 |       39,517 |    353,693 |  4,194,304 |       8.43% |
|      |            |              |            |            |             |
|  1_2 |    147,520 |       33,914 |    181,434 |  4,194,304 |       4.33% |
|      |            |              |            |            |             |
|  2_2 |  1,118,720 |       82,426 |  1,201,146 |  4,194,304 |      28.64% |
|      |            |              |            |            |             |
|  3_2 |    550,720 |       79,244 |    629,964 |  4,194,304 |      15.02% |
|      |            |              |            |            |             |
|  4_2 |  2,163,008 |       48,114 |  2,211,122 |  4,194,304 |      52.72% |
+------+------------+--------------+------------+------------+-------------+
|  All |  4,826,112 |      342,539 |  5,168,651 | 25,165,824 |      20.54% |
+------+------------+--------------+------------+------------+-------------+
```